We have the following discretized x-, y-momentum and continuity equations.

$$\begin{bmatrix} A_x & 0 & B_1 \\ 0 & A_y & B_2 \end{bmatrix} \begin{pmatrix} u_{k+1} \\ v_{k+1} \\ p_{k+1} \end{pmatrix} = b_0 \begin{pmatrix} u_k \\ v_k \end{pmatrix} + b_1 \tag{1}$$

$$\underbrace{\begin{bmatrix} B_1^T & B_2^T \end{bmatrix}}_{B} \begin{pmatrix} u_{k+1} \\ v_{k+1} \end{pmatrix} = b_2 \tag{2}$$

Define

$$b = B^{\dagger} b_2, \qquad P = \text{null}(B), \qquad \begin{pmatrix} u_k \\ v_k \end{pmatrix} = P\alpha_k + b \tag{3}$$

Substitute in Eq. 2 the definition of the vector $\begin{pmatrix} u_k^T & v_k^T \end{pmatrix}^T$ one obtains:

$$\underbrace{BP}_{0} \alpha_{k+1} + b_2 = b_2 \tag{4}$$

hence Eq.2 always holds. Also substitute this definition in Eq. 1 and multiply it on the left with P^T yields:

$$P^T \begin{bmatrix} A_x & 0 \\ 0 & A_y \end{bmatrix} (P\alpha_{k+1} + b) + \underbrace{P^T \begin{bmatrix} B_1 \\ B_2 \end{bmatrix}}_{0} p_{k+1} = P^T \left(b_0 P\alpha_k + b_0 b + b_1 \right) \tag{5}$$

equal to

$$\underbrace{P^T \begin{bmatrix} A_x & 0 \\ 0 & A_y \end{bmatrix} P}_{E_s} \alpha_{k+1} = \underbrace{P^T b_0 P}_{A_s} \alpha_k + \underbrace{P^T \left((b_0 b + b_1) - \begin{bmatrix} A_x & 0 \\ 0 & A_y \end{bmatrix} b \right)}_{B_s w + F_s} \tag{6}$$

with w the axial induction factors.